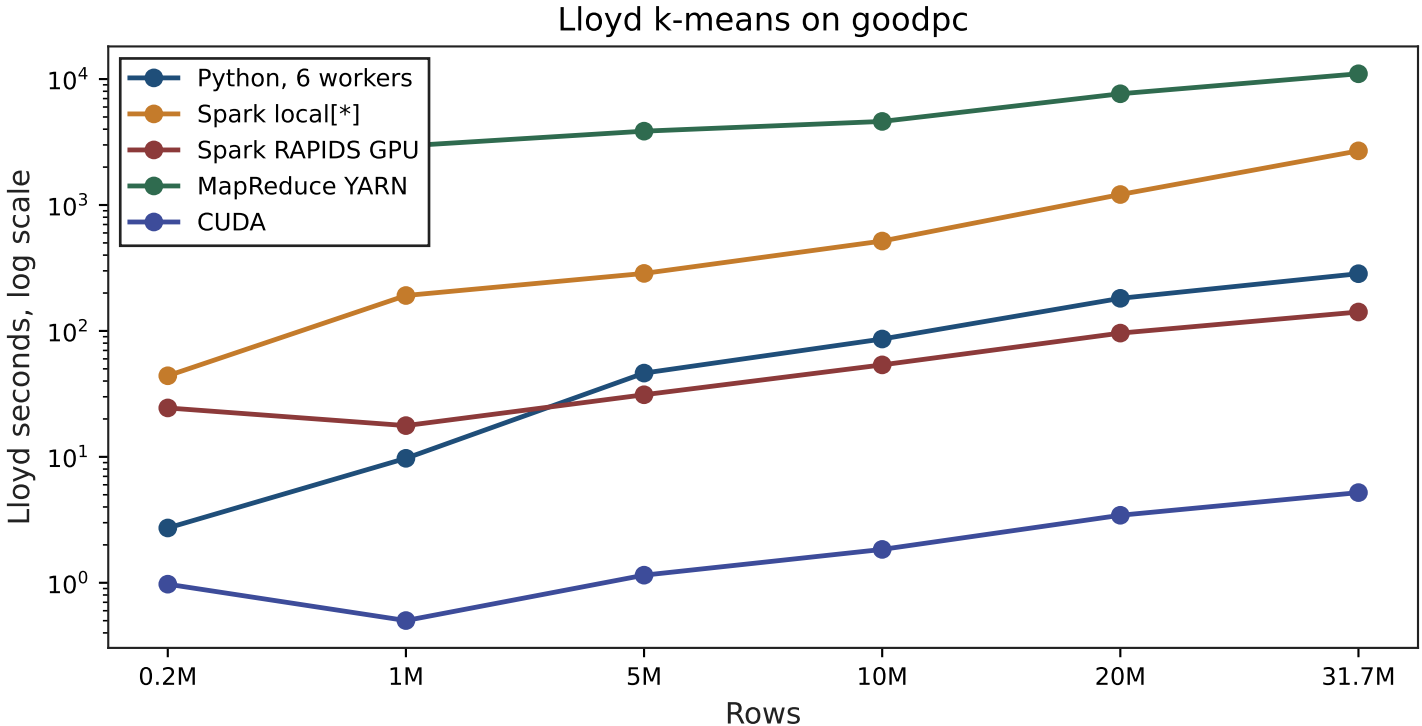

goodpc, Ryzen 5 5600X, 31.89 GB. k=2, stop at 1e-4. Wall starts after the matrix is loaded.

Spark CPU is local[*]. Spark GPU is RAPIDS on the RTX 3070, local[1], one partition.

MapReduce is YARN and excludes HDFS put. CUDA wall excludes the host-to-device copy.

CUDA is faster than Python at every measured size, including 0.2M. No unmeasured crossover is drawn.